

$$n = 4$$

$$p' = (15, 13, 8, 3) \quad w' = (8, 6, 5, 2) \quad C = 18$$

ORDINO GLI INSIEMI IN MODO CHE $\frac{p_s}{w_s}$ SIA DECRESC.

$$1, 875 \quad / \quad 2, 167 \quad , \quad 1, 6 \quad , \quad 1, 5$$

$$p' = (13, 15, 8, 3) \quad w' = (6, 8, 5, 2)$$

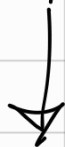
$$M_0 = \{(0, 0)\}$$

$$M_1 = \{(0, 0), (13, 6)\}$$

$$M_2 = \{(0, 0), (13, 6), (15, 8), (28, 14)\}$$

$$M_3 = \{(0, 0), (13, 6), (15, 8), (28, 14), (8, 5), (21, 11), (23, 13)\}$$

$$M_4 = \{(0, 0), (13, 6), \cancel{(15, 8)}, (28, 14), (8, 5), (21, 11), \cancel{(23, 13)}, (3, 2), (16, 8), (18, 10), (31, 16), \cancel{(11, 7)}, (24, 13), \cancel{(26, 15)}\}$$



ORDINO

$$x = (1, 1, 0, 1)$$

$$n = 5$$

$$p' = (15, 13, 9, 8, 3) \quad w' = (8, 6, 8, 5, 2) \quad C = 18$$

$$p' = (13, 15, 8, 3, 9) \quad w' = (6, 8, 5, 2, 8)$$

$$S = \min \left\{ i : \sum_{s=1}^i w_s > c \right\} \quad \bar{c} := c - \sum_{s=1}^{S-1} w_s$$

$$U = \left\lfloor \sum_{s=1}^{S-1} p_s + \bar{c} \frac{p_S}{w_S} \right\rfloor$$

$$\underline{U} = \left\lfloor 13 + 15 + (18 - 6 - 8) \frac{8}{5} \right\rfloor = \left\lfloor \frac{172}{5} \right\rfloor = 34$$

NODO 3:

$$x_3 = 0$$

$$\underline{U} = \left\lfloor 13 + 15 + 3 + (18 - 6 - 8 - 2) \frac{9}{8} \right\rfloor$$

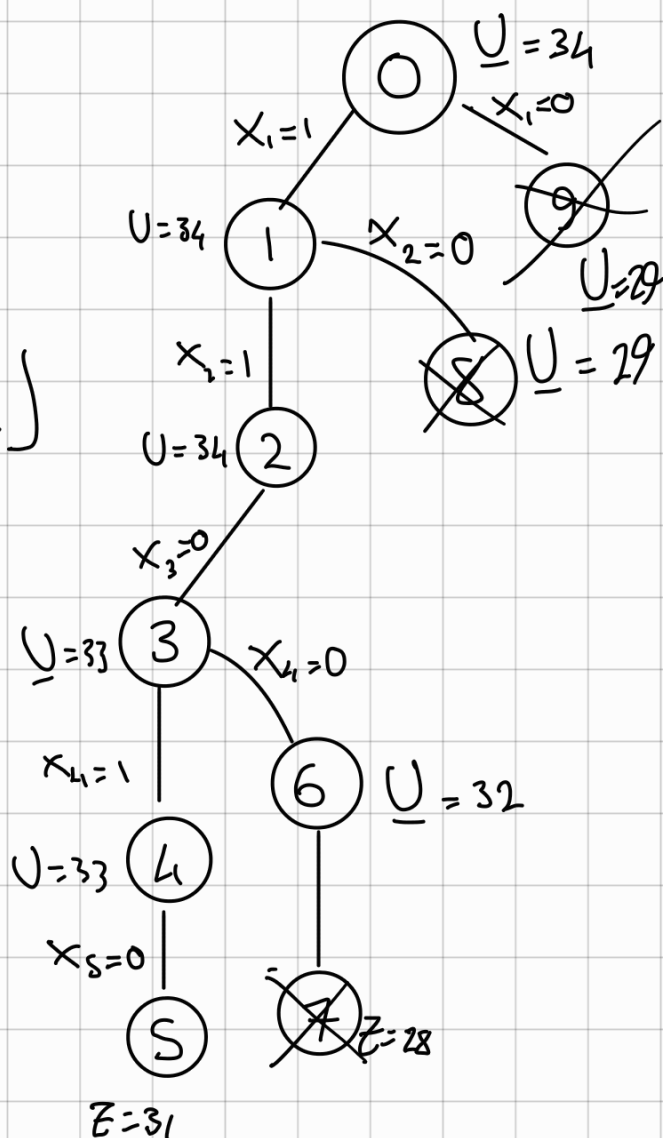
$$= \left\lfloor \frac{133}{4} \right\rfloor = 33$$

NODO 4:

$$U(4) = U(3)$$

$$x_5 = 0 \Rightarrow x = (1, 1, 0, 1, 0)$$

$$\Rightarrow z = 31$$



ESGUO BACKTRACKING:

NODO 6:

$$U = \left\lfloor 13 + 15 + (18 - 6 - 8) \frac{9}{8} \right\rfloor = \left\lfloor \frac{65}{2} \right\rfloor = 32$$

NODO 7:

$$x_5 > 0 \Rightarrow x = (1, 1, 0, 0, 0)$$

$$Z = 28 < 31$$

ESGUO BACKTRACKING:

NODO 8:

$$U = \left\lfloor 13 + 8 + 3 + (18 - 6 - 5 - 2) \frac{9}{8} \right\rfloor = \left\lfloor \frac{237}{8} \right\rfloor = 29$$

$29 < Z$, LO ESCLUDO

NODO 9:

$$U = \left\lfloor 15 + 8 + 3 + (18 - 8 - 5 - 2) \frac{9}{8} \right\rfloor = \left\lfloor \frac{235}{8} \right\rfloor = 29$$

$29 < Z$, LO ESCLUDO

$$\text{Sol. Optima} = 31$$

$$x = (1, 1, 0, 1, 0)$$